KGCS: Zero-Annotation Expert-Knowledge Injection for Object Detection in Aerial Images

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***Abstract*—Aerial object detection is vital for remote sensing applications, yet remains challenged by the prohibitive costs of large-scale data annotation. While current zero-shot detection (ZSD) and open vocabulary detection (OVD) reduce annotation dependency, their reliance on base-class training limits adaptability in complex scenarios. We propose KGCS, a novel zero-annotation framework that achieves accurate detection without any training data through synergistic integration of domain knowledge with foundation models. Our approach introduces two key innovations: a structure-aware dual-path proposal strategy that maintains semantic coherence for composite structures and a tripartite feature dictionary enabling semantic-level alignment beyond conventional category-based matching. Extensive evaluations on DIOR and DOTA datasets demonstrate KGCS's superior performance in both precision and recall, highlighting its potential as an efficient and scalable solution for aerial imagery analysis. The code will be available at https://github.com/HSH55/KGCS**

***Index Terms*—Aerial object detection, expert-knowledge injection, large vision–language model, zero annotation.**

I. Introduction

AERIAL remote sensing object detection is a cornerstone of Earth observation, enabling critical applications such as urban planning, disaster assessment, and environmental monitoring. However, traditional supervised learning methods [1], [2], [3] rely on extensive annotated datasets, incurring high annotation costs and facing scalability challenges due to the scarcity of diverse target categories. Recent efforts attempt to alleviate data bottlenecks through few-shot [4], semisupervised [5], and scattering-aware modeling strategies [6], [7]. While promising, these approaches still require base-class training, handcrafted priors, or supervision on specific structures. To address this, we propose *KGCS*, a zero-annotation detection framework that integrates expert knowledge with general-purpose large models (segment anything model (SAM) [8], contrastive language–image pretraining (CLIP) [9], and generative pretrained transformer-4o (GPT-4o) [10]) to achieve efficient and robust object detection in data-scarce remote sensing scenarios, particularly for common target categories. The hierarchical design of KGCS is illustrated in Fig. 1.

Object detection involves predicting bounding box locations through regression and identifying target categories via classification. Mainstream approaches include single-stage detectors, such as YOLO [11] and SSD [12], which directly predict bounding boxes and class probabilities in a single forward pass, and two-stage detectors, such as Faster R-CNN [13], DETR [14], and Deformable DETR [15], which generate region proposals followed by classification and refinement. In remote sensing, these methods face challenges due to large-scale variations, complex backgrounds, and ambiguous target boundaries [16]. Annotation requires domain expertise, making it costly and difficult to cover diverse targets, particularly rare categories.

To mitigate data dependency, zero-shot detection (ZSD) [17], [18] and open vocabulary detection (OVD) [19], [20], [21], [22] leverage semantic alignment to recognize unseen categories. ZSD uses word embeddings or vision–language models (e.g., CLIP) to model interclass relationships, while OVD extends vocabulary through large-scale image–text pretraining, as demonstrated in GLIP [23], ViLD [24], OWL-ViT [25], and YOLO-World [26]. Few-shot detection methods further reduce data requirements by learning from limited examples [27], [28], [29], with introducing feature reweighting and exploring robust features for satellite imagery [28], [29]. In SAR and ISAR domains, methods such as superpixel-enhanced segmentation [2], [3], [6] and polarimetric-based detection [30], [31], [32] have demonstrated effectiveness for enhancing structure consistency and semantic saliency.

In remote sensing, several adaptations have been proposed to address domain-specific challenges, including CastDet [33], DescReg [34], RS-CLIP [35], and alignment networks enhanced by domain knowledge graphs [18]. These methods incorporate prior knowledge or feature alignment mechanisms to improve generalization; for example, knowledge-graph-based alignment networks leverage remote-sensing semantics for zero-shot scene understanding [18], while CastDet achieves detection with limited examples through generalizable feature learning [33]. Despite these advances, most approaches still rely on base-class training, limiting their adaptability to the

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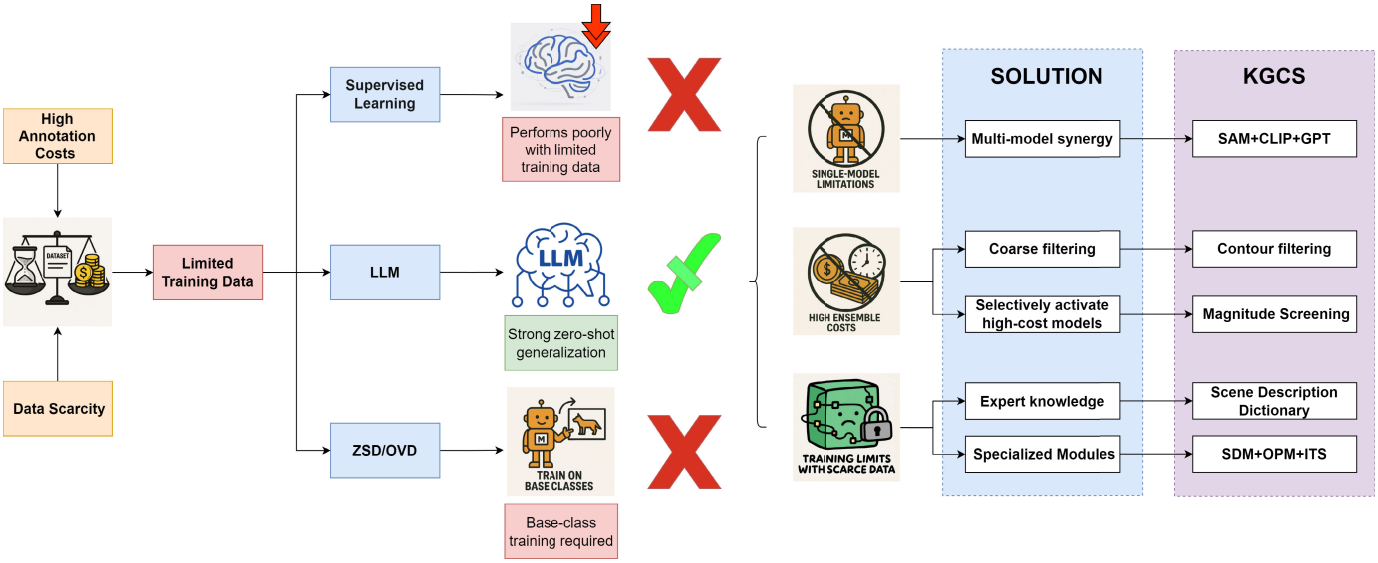


Fig. 1. Hierarchical design logic of KGCS: addressing aerial detection challenges from data scarcity to zero-annotation execution via expert-knowledge-enhanced multimodal collaboration.

diverse and long-tailed distributions common in remote sensing scenarios.

Building on these advances, foundation models like SAM, CLIP, and GPT-4o offer a promising alternative with their powerful zero-shot capabilities. However, their direct application to aerial imagery—whether individually or in naive pipelines—faces fundamental bottlenecks: SAM fragments complex structures, CLIP lacks fine-grained discriminability for overhead views, GPT-4o incurs prohibitive costs for per-proposal reasoning, and all lack inherent domain priors for remote sensing.

To bridge this domain gap, existing solutions primarily follow two paradigms: parameter-based adaptation, which requires fine-tuning on annotated data, or architecture-based fusion, which builds complex multimodal networks. Both, however, remain within the big data learning transfer paradigm, demanding substantial annotations and computational resources.

In this article, we propose KGCS, a novel zero-annotation framework that introduces a paradigm shift by orchestrating frozen foundation models through structured knowledge injection. The core idea is to leverage expert knowledge to guide and synergize these general-purpose models, compensating for their domain-specific shortcomings without any fine-tuning. This is achieved through a meticulously designed pipeline that explicitly addresses the semantic, structural, and instance-level challenges in aerial object detection.

Extensive evaluations under a strict zero-annotation protocol demonstrate that KGCS not only significantly outperforms baseline but also exceeds the performance of base-class-trained generalized zero-shot detection methods, achieving detection performance under more demanding constraints.

II. Proposed Method

The KGCS framework is built upon a structured knowledge injection paradigm, designed specifically to bridge the domain adaptation gap in aerial imagery without requiring annotated data. Unlike conventional approaches that rely on fine-tuning model parameters or constructing complex fusion networks, our core design centers on orchestrating multiple frozen foundation models (SAM, CLIP, and GPT-4o) through a series of explicit, interpretable modules. Each module is strategically devised to address a distinct subset of the challenges outlined in Section I, forming a cohesive pipeline that transforms general-purpose models into domain-specific detectors.

As illustrated in Fig. 2, this design philosophy materializes into a logically structured three-stage pipeline, where each stage progressively refines the representation and understanding of aerial targets:

1) *Scene Description Module (SDM):* The SDM addresses CLIP's limitations in fine-grained discrimination by constructing a discriminative semantic dictionary. It encodes expert knowledge and LLM reasoning into compact textual descriptions that cover attributes such as shape, color, and context, transforming the recognition problem from simple lexical matching to semantic understanding.
2) *Object Proposal Module (OPM):* The OPM tackles SAM's fragmentation issues through a dual-path strategy that adapts to diverse target characteristics. For well-defined objects, it employs geometric validation, while for complex structures, it utilizes mask clustering and adaptive sliding windows to maintain semantic integrity, thereby producing spatially and semantically coherent candidate regions.
3) *Image–Text Similarity Module (ISM):* The ISM enables precise and efficient cross-modal alignment by leveraging the structured outputs from both SDM and OPM. Through adaptive confidence-based screening and distractor suppression mechanisms, it achieves accurate classification while maintaining computational efficiency in zero-annotation settings.

This modular architecture injects domain knowledge at the semantic, structural, and instance levels, forming an automated zero-annotation detection system. The explicit

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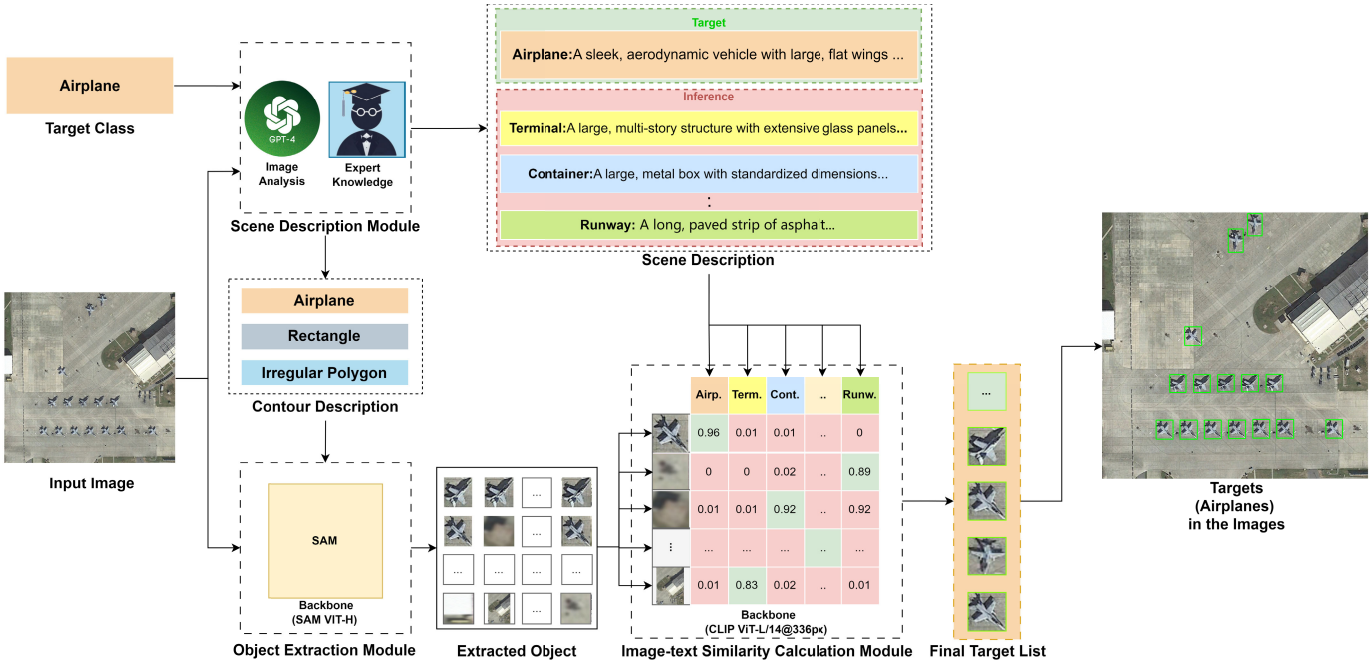


Fig. 2. Workflow of KGCS.

design ensures interpretability while enabling independent component optimization. Subsequent sections detail each module's implementation, demonstrating their collective efficacy in addressing foundation model limitations for remote sensing applications.

A. Scene Description Module

1) Motivation and Problem Definition: The limitations of CLIP in fine-grained classification serve as the direct motivation for constructing the SDM module. As shown in Fig. 3, although CLIP performs well in coarse-grained recognition, it is essentially a *lexical-level matching model*: it classifies by computing similarity between images and individual keywords or phrases, lacking the ability to understand complex semantic relationships between words. Consequently, when different categories (e.g., train stations versus residential buildings) are visually similar, CLIP fails to perform deep semantic reasoning, leading to classification confusion.

2) Core Concept and Module Function: Given the inability to modify CLIP's inherent matching mechanism, the core concept of the SDM module is: within CLIP's lexical-level matching framework, to simulate the effect of semantic-level matching by constructing a highly discriminative description dictionary optimized through prior knowledge.

Specifically, SDM operates through three coordinated mechanisms.

1) *Decomposition and Refinement:* SDM parses complex semantic concepts (e.g., *train station*) into discriminative visual attributes (e.g., *long platform* and *parallel railway tracks*) that align with CLIP's recognition capabilities.
2) *Exclusion-Oriented Design:* The dictionary construction follows a *quality-over-quantity* principle, incorporating distractor descriptions to actively exclude semantically ambiguous terms from CLIP's feature space.

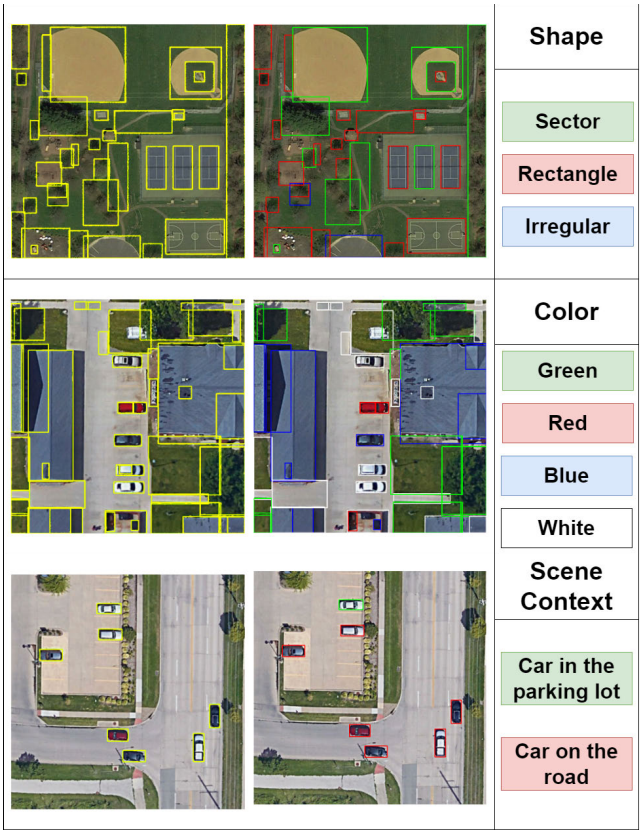


Fig. 3. CLIP-based classification results. Left: filtered proposals. Right: CLIP predictions. From top to bottom: shape, color, and context.

3) *Hierarchical Matching:* Through multilevel feature descriptions (contour, color, and context), SDM orchestrates focused lexical matching operations whose outputs are integrated in downstream modules.

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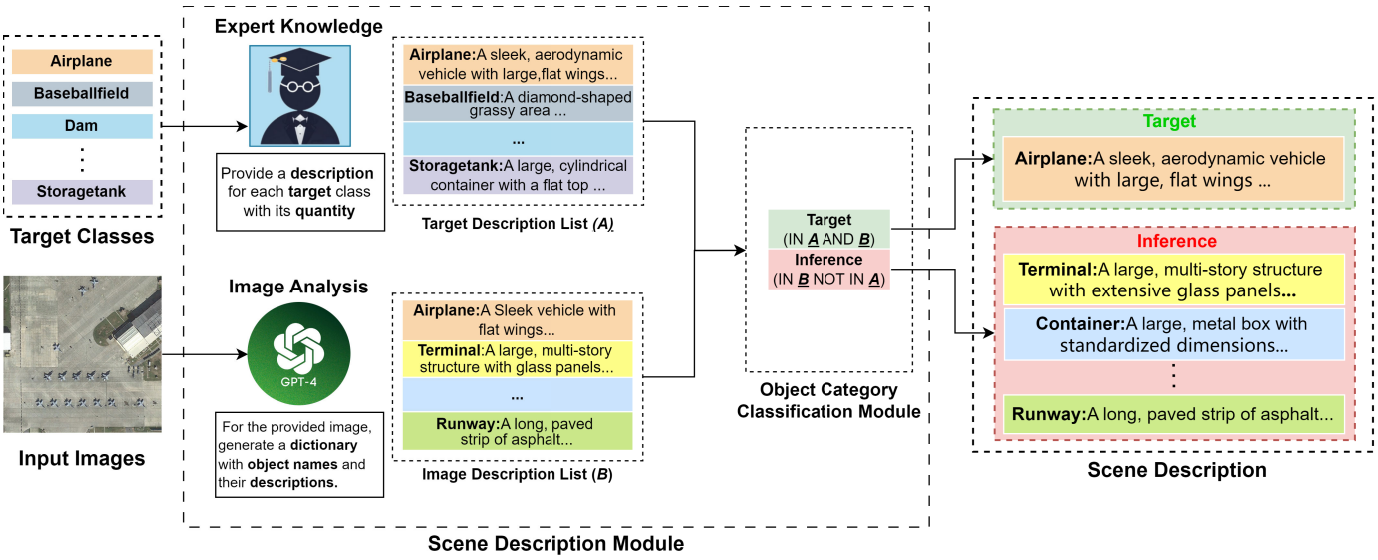


Fig. 4. Overall architecture of the SDM. The module integrates expert knowledge with large language model reasoning to build highly discriminative description dictionaries for assisting CLIP in achieving more accurate classification.

Thus, rather than augmenting CLIP's inherent capabilities, SDM enhances its decision-making reliability through structured prior knowledge orchestration.

3) Hierarchical Dictionary Construction: SDM systematically encodes prior knowledge at different levels by constructing three functionally complementary specialized dictionaries.

1) *Target Semantic Description Dictionary*

$$\mathcal{D}_{\text{target}} = \left\{ (c_i, d_i) \;\middle|\; c_i \in \mathcal{C}_{\text{target}} \right\}. \tag{1}$$

This is the core output of SDM, generating a composite semantic description d_i for each target category c_i, comprehensively covering its typical *shape, color, and contextual* features. All target descriptions are defined by domain experts to ensure accuracy and authority.

2) *Distractor Description Dictionary*

$$\mathcal{D}_{\text{distractor}} = \left\{ (c_j, d_j) \;\middle|\; c_j \in \mathcal{C}_{\text{distractor}} \right\}. \tag{2}$$

As a key discriminative enhancement component, this dictionary aims to exclude common background elements or easily confusable objects that co-occur with targets.

3) *Contour Feature Dictionary*
This dictionary focuses on geometric boundary prototype descriptions, providing geometric priors for the structure-aware preliminary screening in the OPM module, aiding in distinguishing true target contours from noise.

4) Dictionary Construction Strategy and Design Principles: The dictionary construction in this framework adheres to the following key design principles to ensure discriminativity, compactness, and generalizability:

Representative Distractor Selection Strategy: Theoretically, dynamically generating distractors for each specific scenario is ideal, but this is infeasible in a zero-annotation framework and leads to dictionary bloat. Therefore, we adopt a compromise strategy: leveraging GPT-4o's powerful scene understanding and summarization capability to perform a one-time analysis on numerous aerial scenes, extracting cross-scenario, cross-category common high-frequency distractors (e.g., *woodland*, *rooftop*, *water body*, and *shadow*). This aims to build a unified, compact, and highly representative distractor description library.

Dictionary Size Optimization: To balance CLIP's discriminative power and retrieval efficiency, we experimentally restrict the size of the final fused dictionary to *4–5 semantic entries*. A typical configuration is: *1 target description* + 3–*4 common distractor descriptions*. This size introduces sufficient discriminative information while preventing a significant drop in recall due to an excessive number of categories.

Targeted Distractor Exclusion Mechanism: Distractor descriptions are applied in two modes based on target boundary characteristics.

1) For *Fixed Boundary Targets* (e.g., *basketball court*): After OPM performs contour-based preliminary screening, distractor descriptions (e.g., *square rooftop*) primarily provide auxiliary details like color and material for fine-grained distinction in the ISM module.
2) For *Nonfixed Boundary Targets:* Distractor descriptions (e.g., *golffield* and *expressway-service-area*) are used in the ISM module for direct semantic-level comparison, effectively excluding background interference by emphasizing the *distinctive features* that differentiate the target from these common distractors.

5) Dictionary Construction Pipeline: Building upon these design principles, SDM constructs the aforementioned dictionaries through an efficient pipeline as shown in Fig. 4.

1) *Expert Knowledge Injection:* Manually define the discriminative features for each target category and enumerate typical distractors. Expert descriptions focus on the most distinctive visual features, avoiding semantically ambiguous or interclass similar descriptions.

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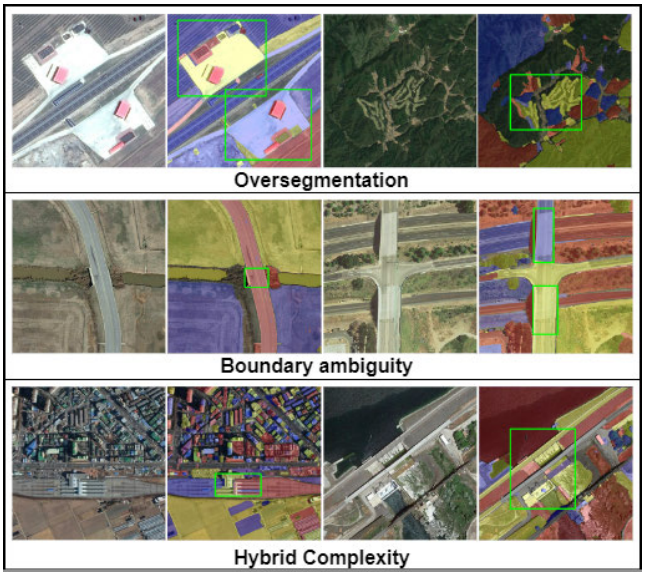


Fig. 5. Illustration of typical segmentation issues encountered in boundary-ambiguous or compositional targets.

2) *One-Shot Scene Parsing:* Utilize a large language model (GPT-4o) to perform a single inference on the entire image, extracting scene-level entities and contextual information. The prompt template is: "Describe all possible targets in this aerial image and the features of their easily confusable distractors."
3) *Dictionary Fusion and Optimization:* Integrate the expert-defined features with the LLM output

$$\mathcal{D}_{\text{final}} = \mathcal{D}_{\text{expert}} \cup \text{Filter}\left(\mathcal{D}_{\text{LLM}}\right). \tag{3}$$

The filter operation performs strict semantic alignment, redundancy elimination, and removal of low-discriminability distractors, with particular emphasis on excluding descriptions that might confuse CLIP's feature space, ensuring each entry in the final dictionary has sufficient semantic distinctiveness.

To ensure system efficiency, only 3–5 representative images per target category are used to initiate this process. The output of SDM—the constructed hierarchical description dictionary—serves as structured semantic knowledge and is directly provided to the ISM module for subsequent precise matching.

B. Object Proposal Module

1) Motivation and Problem Definition: In the context of zero-annotation remote sensing object detection, a fundamental challenge is how to automatically and efficiently generate high-quality object candidate regions (i.e., proposals) without relying on any manually annotated data. Existing fully supervised or weakly supervised methods are unsuitable under this constraint, while traditional unsupervised approaches (e.g., selective search) often prove inadequate when faced with the complex backgrounds, extreme scale variations, and diverse target morphologies characteristic of remote sensing imagery. Consequently, there is a pressing need for an intelligent proposal generation mechanism that can leverage the general capabilities of advanced foundation models while overcoming their domain-specific limitations.

To address this, we propose the OPM. The core objective of this module is to build a proposal generator that is *completely free of training and annotation*, providing accurate and spatially compact candidate regions for subsequent zero-shot classification.

We select *SAM* as the backbone of OPM, primarily for the following two reasons.

1) *Potent Zero-Shot Generalization:* Trained on a massive dataset, SAM possesses a deep understanding of general visual concepts and can perform high-quality segmentation on unseen objects and scenes. This aligns perfectly with our requirements of zero annotation and strong generalization.
2) *Rich Mask Output:* SAM can generate dense, pixel-level masks for multiple potential objects or regions within an image, providing abundant raw material from which candidate detection boxes can be filtered and reconstructed.

However, as a model designed for general-purpose scenes, SAM's output cannot be used directly as high-quality proposals for remote sensing detection. As illustrated in Fig. 5, applying SAM directly to aerial imagery reveals several fundamental issues.

1) *Over-Segmentation of Composite Targets:* Macrotargets consisting of multiple functional subunits, such as harbors and service areas, are often fragmented by SAM into numerous disconnected masks, thereby compromising their semantic and structural integrity.
2) *Insufficient Delineation of Targets With Ambiguous Boundaries:* Objects like bridges, industrial parks, and storage yards often lack well-defined, closed contours and exhibit gradual transitions to the background. This characteristic leads SAM to produce incomplete or inaccurate segmentations.
3) *Incomplete Coverage in Heterogeneous Scenes:* In complex scenes containing both distinct and amorphous structures (e.g., a train station with platforms and rail yards), a single forward pass of SAM often fails to capture all semantically relevant regions, resulting in proposals that miss critical components.

These issues indicate that SAM's raw output is generally unsuitable for direct use as detection proposals. Therefore, a dedicated postprocessing mechanism is required to interpret and reassemble the underlying macrostructure of targets from these primitive segments.

2) Limitations of Conventional Approaches: Existing methods for generating region proposals under low-data constraints are hampered by several critical shortcomings.

1) *Morphological Postprocessing:* These operations are inherently parameter-sensitive and lack semantic guidance. Their primary limitation is the inability to robustly distinguish between the boundaries of adjacent objects and the internal structural gaps within a single, complex target. Consequently, they often either fail to separate

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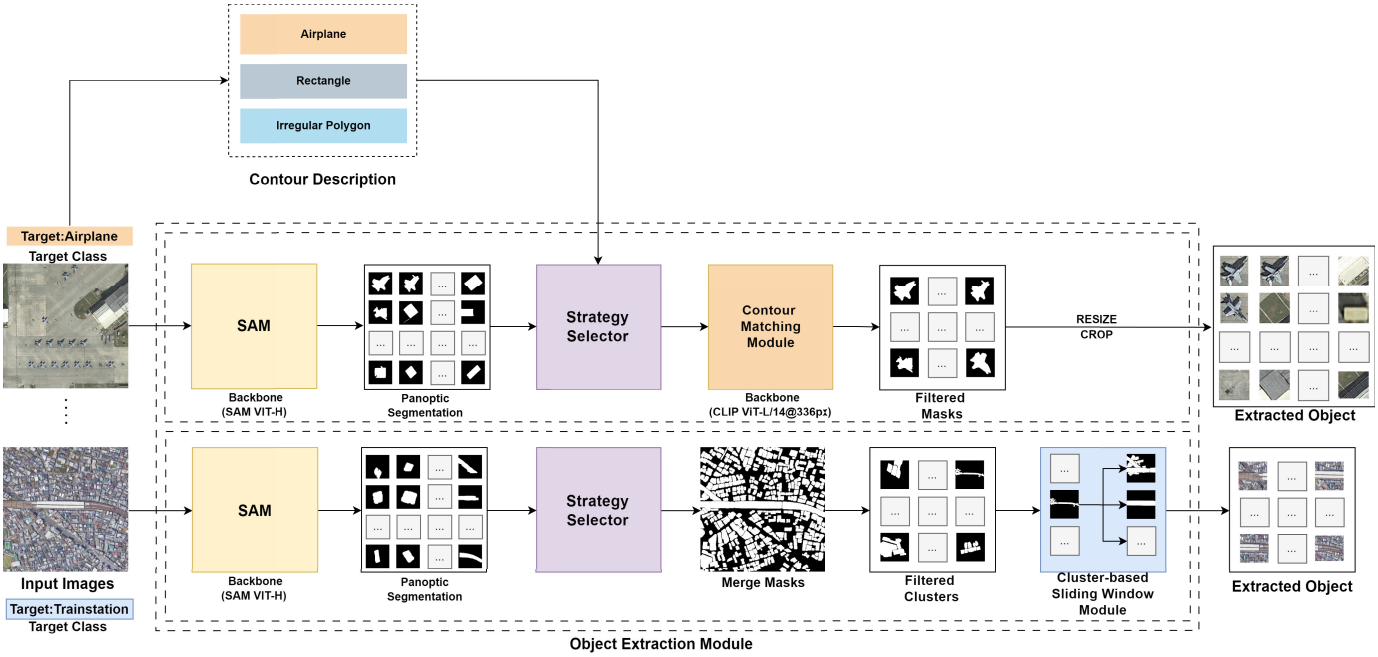


Fig. 6. Workflow of the proposed OPM. The left path handles contour-clear targets with direct validation, while the right path addresses boundary-ambiguous or compositional structures via structure-guided clustering and adaptive sliding windows.

adjoining objects or erroneously break apart a coherent structure, especially at scale.

2) *Heuristic-Based Mask Merging:* Strategies that rely on simple, low-level metrics such as Intersection over Union (IoU) or mask area operate without an understanding of higher-order geometric or functional relationships. This frequently results in semantically incoherent mergers—either combining unrelated background regions with the target or failing to merge fragments that belong to the same logical entity.
3) *Exhaustive Multiscale Sliding Windows:* While exhaustive in coverage, this strategy is computationally prohibitive. To account for unknown target sizes, it must perform dense sampling across the entire image, generating an overwhelming number of candidate boxes (often $> 10^4$ per image). This leads to severe redundancy, a high false-positive rate, and places an unsustainable processing burden on downstream stages.

The fundamental flaw common to all these approaches is their *lack of domain-aware structural reasoning* and an *effective adaptive mechanism*, rendering them ill-suited to handle the immense scale variation and target diversity inherent in zero-annotation remote sensing.

3) Structure-Aware Dual-Path Proposal Strategy: To overcome the above limitations, we propose a structure-aware OPM, as shown in Fig. 6. This module adopts a dual-path design based on boundary characteristics, with dedicated proposal generation strategies for each type.

a) Path 1: contour-clear targets: For geometrically compact and visually distinct targets (e.g., airplanes, ships, and tanks), SAM usually outputs high-quality, complete masks. We directly extract valid candidate boxes using the following procedure.

1) *Mask Generation:* From SAM output, obtain mask set $\mathcal{M} = \{m_1, m_2, \ldots, m_n\}$.
2) *Bounding Box Proposal:* Convert each mask to a bounding box, $B = \{b_i = \mathrm{BBox}(m_i)\}$.
3) *CLIP-Based Shape Validation:*

$$B_{\text{valid}} = \left\{ b_i \,\middle|\, \Psi_{\text{CLIP}}\left(\text{Crop}\left(I, b_i\right), d_{\text{contour}}\right) > \tau \right\} \tag{4}$$

where d_{contour} is the textual geometric descriptor and τ is a threshold for filtering.

This path yields compact, low-redundancy proposals for structurally well-defined objects.

b) Path 2: boundary-ambiguous or compositional targets: For targets with unclear or nonfixed boundaries—such as bridges, train stations, and storage facilities—we introduce a two-stage structure-aware proposal generation process.

Stage 1: Structure-Guided Mask Clustering

SAM output often includes fragmented masks that represent parts of a single structure. We apply geometric clustering based on spatial connectivity to group relevant masks

$$\mathcal{C} = \text{Cluster}\left(\mathcal{M}_{\text{valid}}\right). \tag{5}$$

Each cluster $c_k \in \mathcal{C}$ forms a structure-consistent candidate region, preserving the spatial and functional integrity of composite objects.

Stage 2: Adaptive Sliding Window Generation

For each cluster, we generate proposal windows using a structure-guided adaptive sliding strategy.

Window Size:

$$S = L \cdot W \tag{6}$$

where L and W are the long and short sides of the cluster's bounding box, respectively.

Step Size:

$$\Delta = \frac{S}{3} \cdot \left(1 - \frac{N_{\text{cluster}}}{N_{\max}}\right) \tag{7}$$

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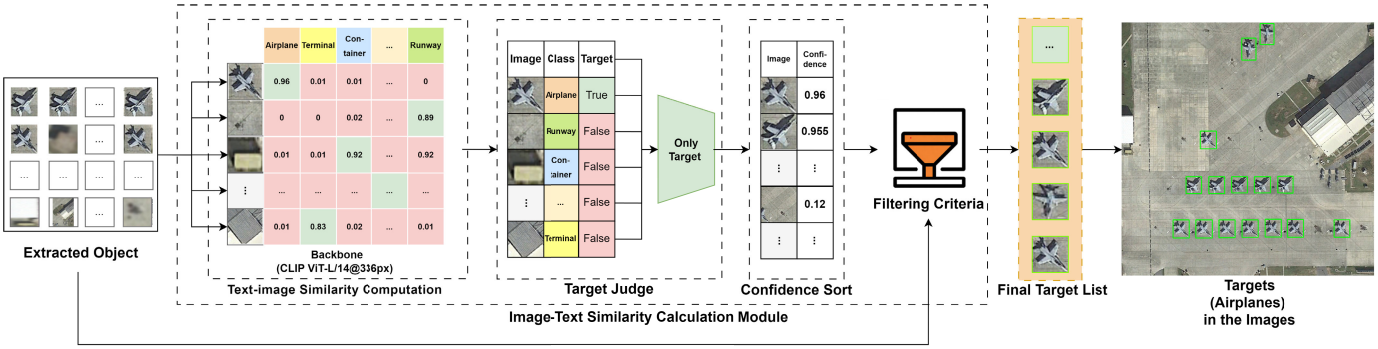


Fig. 7. Workflow of the Image-Text Similarity Module (ISM). The module includes four stages: Text-image Similarity Computation, Target Judge, Confidence Sort, and Filtering Criteria, where proposals are matched with semantic descriptions, assigned to the most relevant category, ranked by similarity confidence, and finally filtered using dynamic threshold calibration and boundary-type-aware ratio filtering to obtain the final detection results.

where N_{cluster} is the number of masks in the cluster and $N_{\max}$ is the maximum across all clusters.

Multiscale Views:

$$W_{\text{standard}} = S, \quad W_{\text{large}} = 1.05\ S, \quad W_{\text{small}} = 0.95\ S. \tag{8}$$

These variants capture structural uncertainty.

1) W_{large} includes more context to handle missing or disconnected parts.
2) W_{small} focuses on precise localization by excluding redundant boundaries.

This approach produces spatially focused and semantically meaningful proposals, especially for ambiguous or overlapping targets.

C. Image–Text Similarity Module

1) Module Objective and Methodological Design: As illustrated in Fig. 7, the ISM serves as the final stage for classification and localization within the KGCS framework. Its primary objective is to accurately assign semantic labels from the SDM dictionary to the candidate regions generated by OPM under a strict zero-annotation protocol. To overcome the inherent limitations of uniform selection strategies, the ISM adopts a *two-stage adaptive filtering mechanism*.

Considering that in zero-annotation scenarios, the confidence scores produced by the CLIP model—without task-specific fine-tuning—may lack significant interclass distinction and variability, relying solely on score distribution for dynamic separation can be unstable. Therefore, the core methodological innovation of this module lies in its *calibration-safeguard two-stage design*, which consists of two sequentially applied strategies.

1) *Confidence-Guided Dynamic Threshold Calibration:* First, an adaptive dynamic threshold is calculated based on the statistical distribution of confidence scores within each category, attempting an initial, data-driven quality screening.
2) *Boundary-Aware Ratio Filtering Safeguard:* Subsequently, leveraging the structural prior from OPM, differentiated fixed-ratio filtering is applied to the two target types: "contour-clear" and "boundary-ambiguous." This step serves as a crucial safeguard, ensuring that even in extreme cases where confidence score fluctuation is minimal, the system can maintain a controllable and reliable selection lower bound based on the structural characteristics of targets, effectively preventing performance collapse due to dynamic threshold failure.

This design prioritizes calibration using intrinsic data distribution, with a domain knowledge (boundary type)-driven ratio rule as a safety net, collectively ensuring a robust tradeoff between recall and precision in complex scenarios.

2) Cross-Modal Matching and Confidence Assignment: The module first computes the semantic alignment between each visual proposal and all target descriptions in the SDM dictionary.

a) Feature encoding: For each target category description d_k from SDM and each visual region b_i from OPM, corresponding embedding vectors are generated

$$\mathbf{t}_k = f_{\text{text}}(d_k), \quad \mathbf{v}_i = f_{\text{vision}}(\mathcal{R}(I, b_i)). \tag{9}$$

b) Confidence score calculation: The final confidence score for a proposal is defined as its maximum cosine similarity with any target category description, while its category label is also recorded

$$s_i = \max_{k \in \mathcal{C}_{\text{target}}} \left(\frac{\langle \mathbf{v}_i, \mathbf{t}_k \rangle}{\|\mathbf{v}_i\|_2 \cdot \|\mathbf{t}_k\|_2} \right)$$
$$l_i = \arg \max_{k \in \mathcal{C}_{\text{target}}} \left(\frac{\langle \mathbf{v}_i, \mathbf{t}_k \rangle}{\|\mathbf{v}_i\|_2 \cdot \|\mathbf{t}_k\|_2} \right). \tag{10}$$

3) Stage 1: Confidence-Distribution-Guided Dynamic Threshold Calibration: In this stage, ISM performs internal calibration separately on the two sets of proposals from OPM ($\mathcal{B}_{\text{clear}}$ and $\mathcal{B}_{\text{ambiguous}}$). This step aims to preliminarily filter out obvious outlier low-score proposals that deviate from the main distribution, utilizing the distribution characteristics of the confidence scores within each target type.

For each set to be processed $\mathcal{B} \in \{\mathcal{B}_{\text{clear}}, \mathcal{B}_{\text{ambiguous}}\}$, calculate the mean μ and standard deviation σ of its confidence scores s_i

$$\mu = \frac{1}{|\mathcal{B}|} \sum_{b_i \in \mathcal{B}} s_i, \quad \sigma = \sqrt{\frac{1}{|\mathcal{B}|} \sum_{b_i \in \mathcal{B}} (s_i - \mu)^2}. \tag{11}$$

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Subsequently, a dynamic threshold T_{dynamic} is calculated based on this distribution

$$T_{\text{dynamic}} = \mu - \gamma \cdot \sigma. \tag{12}$$

Here, γ is a factor controlling the strictness of filtering. Considering the inherent uncertainty of ambiguous targets, a slightly larger γ value can be applied to them. The original set is preliminarily screened using this threshold

$$\mathcal{B}_{\text{calibrated}} = \left\{ b_i \in \mathcal{B} \mid s_i > T_{\text{dynamic}} \right\}. \tag{13}$$

4) Stage 2: Differentiated Ratio Filtering Safeguard Based on the Boundary Type: Following dynamic threshold calibration, the ISM executes the crucial safeguarding filtering step. This step directly addresses the potential issue of CLIP scores having a flat distribution in zero-shot scenarios, which could render the dynamic threshold ineffective for separation. We leverage the more robust structural prior provided by OPM—target boundary type—and apply different retention ratios to the two calibrated sets.

1) *For Contour-Clear Targets ($\mathcal{B}^{\text{cal}}_{\text{clear}}$):* The matching results for these targets are generally more reliable. To prioritize high recall, a *higher* retention ratio ρ_c is applied (e.g., retaining the top R_c% of proposals in the calibrated set)

$$\mathcal{B}'_{\text{clear}} = \text{TopRatio}\left(\mathcal{B}^{\text{cal}}_{\text{clear}}, s_i, \rho_c\right). \tag{14}$$

2) *For Boundary-Ambiguous/Composite Targets ($\mathcal{B}^{\text{cal}}_{\text{ambiguous}}$):* These targets have high uncertainty and are a major source of false positives. To strictly control the precision of the system output, a *lower* retention ratio ρ_a is applied (e.g., retaining the top R_a%, where $R_a < R_c$)

$$\mathcal{B}'_{\text{ambiguous}} = \text{TopRatio}\left(\mathcal{B}^{\text{cal}}_{\text{ambiguous}}, s_i, \rho_a\right). \tag{15}$$

Here, $\text{TopRatio}(\mathcal{B}, s, \rho)$ selects the top $\rho \times 100\%$ of proposals from set $\mathcal{B}$ when ranked in descending order by confidence score s. The ratio parameters ρ_c and ρ_a constitute a stable lower bound for system performance and are important hyperparameters in the zero-annotation framework.

Finally, the outputs from the two filtered categories are merged

$$\mathcal{B}_{\text{final}} = \mathcal{B}'_{\text{clear}} \cup \mathcal{B}'_{\text{ambiguous}}. \tag{16}$$

III. Experiments

A. Overview of Experiments

To validate the effectiveness of KGCS under *strict zero-annotation conditions*, we design a series of experiments with the core objective of evaluating how *structured expert-knowledge injection* compensates for the inherent limitations of general-purpose foundation models (CLIP and SAM) in the aerial domain. The experiments are not intended to compete with traditional fully supervised methods for absolute accuracy, but instead focus on demonstrating the performance gains and feasibility achieved through knowledge-guided model collaboration, *without any annotated data or model fine-tuning*. The following experiments will systematically present the efficacy of the KGCS framework, from settings, task definition, and datasets to specific results.

B. Experimental Setting

We conduct comprehensive experiments on two widely used remote sensing datasets: DOTA [36] and DIOR [37].

1) The *DOTA* dataset contains 2806 aerial images with resolutions ranging from 800 to 4000 pixels, annotated with 15 common object categories.
2) The *DIOR* dataset comprises 23 463 images at a uniform resolution of 800 × 800 pixels, containing 1 92 472 object instances across 20 categories.

Following DescReg [34] for consistency, we resize all DOTA images to 800 × 800 pixels. This results in 5863 test images for DIOR and 6259 for DOTA. This standardization ensures fair and comparable evaluation across both datasets.

C. Task Formulation

Our experiments are specifically designed to evaluate knowledge transfer in zero-annotation detection, rather than benchmarking base model capacity. Since zero-annotation aerial detection lacks established benchmarks, we position KGCS within the closest related paradigm—generalized ZSD (GZSD)—while imposing a stricter protocol: *complete elimination of base-class training*.

Unlike traditional ZSD and OVD methods that rely on thousands of seen-class bounding boxes or large-scale image–text alignment, KGCS performs inference solely through expert-knowledge injection and frozen foundation models, without any gradient updates on aerial data. This setting better reflects real-world scenarios where annotation is infeasible, and data are scarce.

The core challenges in this strict zero-annotation setting present significant obstacles that distinguish our work from conventional zero-shot learning paradigms.

1) *Zero Base-Class Training*—The complete absence of aerial bounding-box annotations for model updates necessitates reliance solely on frozen foundation models and external knowledge injection, eliminating any gradient-based adaptation to the target domain.
2) *Composite-Structure Ambiguity*—Complex aerial targets such as ports and industrial complexes exhibit fragmented visual patterns in overhead imagery. Foundation models like SAM tend to over-segment these structures, requiring advanced reasoning to maintain semantic coherence across disconnected components.
3) *Overhead-View Bias*—Web-pretrained vision–language models like CLIP lack specialized knowledge of the unique geometric transformations, scale variations, and texture patterns characteristic of aerial perspectives. This domain gap leads to fundamental limitations in fine-grained discrimination of overhead objects.
4) *Resource Constraints*—All experiments are conducted under strict computational limitations: single GPU deployment without token retraining or model fine-tuning, ensuring practical feasibility for real-world applications while maintaining the zero-annotation principle.

Consequently, our evaluation strategy deliberately emphasizes interpretable performance gains achieved through

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TABLE I

SEEN AND UNSEEN CATEGORIES FOR THE DIOR AND DOTA DATASETS

Dataset	Categories	
	Seen	Unseen
DIOR	airplane, baseballfield, bridge, chimney, dam, Expressway-Service-area, Expressway-toll-station, golffield,harbor, overpass,ship, stadium, storagetank, tenniscourt, trainstation, vehicle	airport, basketballcourt, groundtrackfield, windmill
DOTA	plane, ship, storage-tank, baseballdiamond, basketball-court, ground-track-field, harbor, bridge, large-vehicle, small-vehicle, roundabout	tennis-court, helicopter, soccer-ballfield, swimming-pool

TABLE II

FIXED AND NONFIXED BOUNDARY CATEGORIES FOR THE DIOR AND DOTA DATASETS

Dataset	Categories	
	Fixed	Non-fixed
DIOR	airplane, baseballfield, bridge, chimney, toll-station, groundtrackfield, ship, storagetank, tenniscourt, vehicle, windmill	airport, basketballcourt, dam, Servicearea, golffield, harbor, overpass, stadium, trainstation
DOTA	Baseballdiamond, Basketballcourt, Ground-track-field, Helicopter, Large-vehicle, Plane, Ship, Small-vehicle, Soccer-ball-field, Tennis-court	Bridge, Container-crane, Harbor, Roundabout, Swimming-pool

structured knowledge injection, rather than pursuing absolute accuracy benchmarks against fully supervised methods. This approach better reflects the practical value of our framework in scenarios where annotation is infeasible, while providing transparent insights into how expert knowledge compensates for the limitations of general-purpose foundation models in specialized domains.

D. Dataset Details

The details of the two remote sensing datasets are as follows.

1) Image Cropping: To evaluate the effectiveness of our methods in target detection across different categories, we reorganize the test and validation images along with their annotations. For the DOTA dataset, we resize and crop the images to 800 × 800 pixels, ensuring maximum coverage of the annotated areas. For both datasets, we use the images as-is for GZSD experiments, following the approach in DescReg [34].

2) Seen/Unseen Split: For DIOR, we follow RRFS [38] to split the 20 classes into 16 seen and 4 unseen classes. For DOTA, the resulting splits are 11 seen and 4 unseen. These class splits are shown in Table I.

3) Fixed/Nonfixed Boundary Split: For DIOR, we split the 20 classes into 11 fixed and 9 nonfixed classes. For DOTA, the resulting splits are 10 fixed and 5 nonfixed. These class splits are shown in Table II.

E. Experimental Results and Analysis

1) Performance Comparison on the DIOR Dataset: The comparative results on the DIOR dataset under GZSD settings are presented in Table III, while Fig. 8 provides visual analysis of performance across different boundary types. KGCS demonstrates substantial improvements over existing ZSD approaches without requiring any training data or fine-tuning.

KGCS demonstrates remarkable and consistent performance across all evaluation protocols on the DIOR dataset, as detailed in Table III.

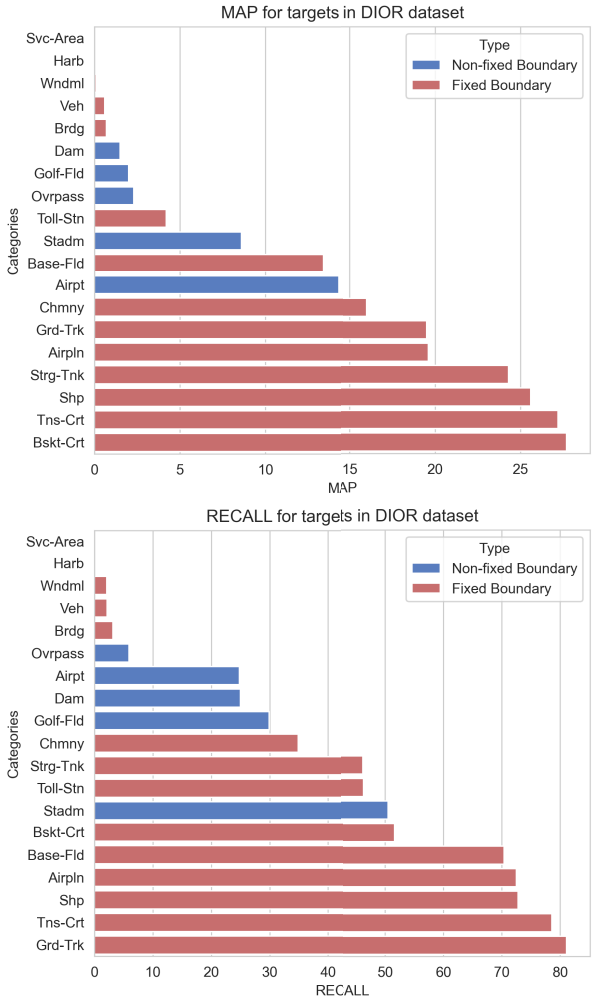


Fig. 8. MAP and RECALL values for nonfixed and fixed boundary targets in the DIOR dataset, sorted by value.

TABLE III

COMPARISON WITH METHODS UNDER GZSD SETTINGS ON THE DIOR DATASET

Method	GZSD	
	Recall@100	mAP
BLC (Zheng et al., 2020)	-	0.4
SU (Hayat et al., 2020)	-	2.9
RRFS (Huang et al., 2022)	-	3.4
V2S† (Khandelwal et al., 2023)	15.8	1.4
RRFS† (Huang et al., 2022)	19.9	2.8
ContrastZSD† (Yan et al., 2022)	25.9	3.9
DescReg (Zang et al., 2024)	34.3	7.9
KGCS val novel(ours)	**42.9**	**8.4**
KGCS test novel(ours)	**35.2**	**11.4**
KGCS test all (ours)	**48.8**	**15.4**

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TABLE IV
COMPARISON WITH METHODS UNDER GZSD SETTINGS ON THE DOTA DATASET

Method	GZSD	
	RE@100	mAP
RRFS	14.2	2.2
ContrastZSD	12.2	2.8
DescReg	29.9	4.7
KGCS novel(ours)	**52.6**	**5.3**
KGCS all(ours)	**51.0**	**5.4**

1) *Validation Set (Novel Classes):* KGCS achieves 42.9% Recall@100 and 8.4% mAP. This result not only outperforms all conventional ZSD baselines but, more importantly, surpasses the recent state-of-the-art DescReg (34.3% Recall@100, 7.9% mAP). This significant margin validates our framework's superior knowledge transfer capability under strict zero-annotation constraints.
2) *Test Set (Novel Classes):* On the balanced test subset, KGCS maintains robust performance with 35.2% Recall@100 and 11.4% mAP. The strong recall indicates effective detection coverage for novel categories, while the competitive mAP demonstrates precise localization even without any task-specific training.
3) *Test Set (All Classes):* When evaluated across all 20 categories, KGCS reaches 48.8% Recall@100 and 15.4% mAP. This comprehensive performance confirms that our expert-knowledge injection framework benefits both seen and unseen categories, enabling effective generalized zero-shot detection in practical aerial scenarios.

The consistent superiority of KGCS across these evaluations strongly validates the efficacy of our structured knowledge injection paradigm. Notably, KGCS outperforms DescReg by 8.6% in recall and 0.5% in mAP on novel classes, despite DescReg requiring base-class training. This performance advantage highlights how structured domain knowledge can compensate for the lack of annotated data, offering a more practical solution than conventional zero-shot learning techniques that rely solely on semantic embeddings or limited visual-text alignment.

2) Performance Comparison on the DOTA Dataset: To further assess the robustness and generalizability of KGCS, we extend our evaluation to the DOTA dataset. Following the established protocol, we concentrate on the validation set due to the absence of test labels. To ensure balanced evaluation, we adjust the proportion of small-vehicle samples, which account for over 50% of total detections.

Our method demonstrates substantial improvements on the DOTA dataset across different evaluation scopes.

1) *Novel Classes:* KGCS achieves 52.6% Recall@100 and 5.3% mAP for novel categories, representing a significant improvement of 22.7% in recall and 0.6% in mAP compared to the strongest baseline DescReg.
2) *All Classes:* When evaluating across all 15 categories, KGCS attains 51.0% Recall@100 and 5.4% mAP, demonstrating an overall enhancement of 21.1% in recall and 0.7% in mAP.

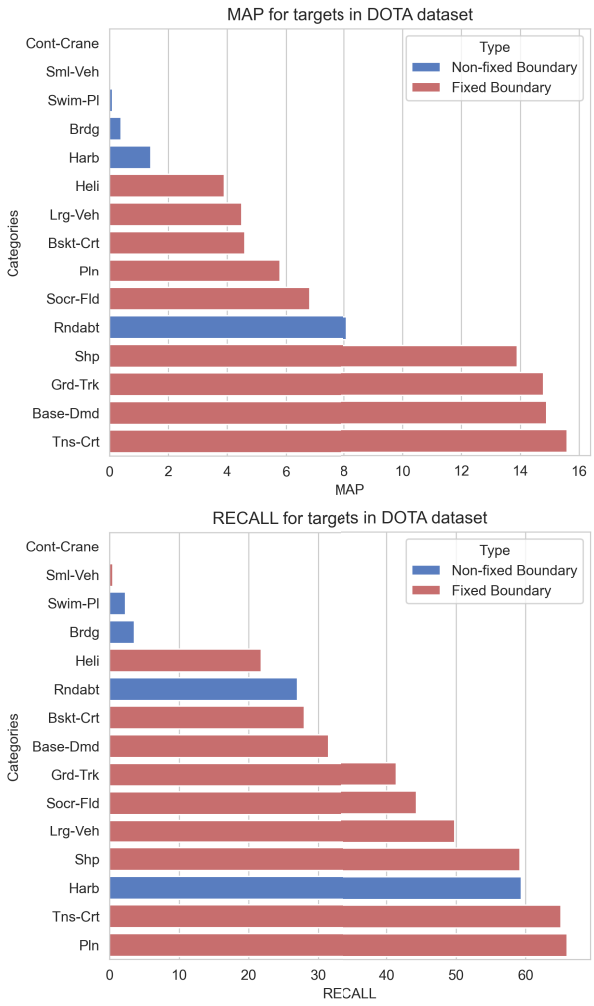


Fig. 9. MAP and RECALL values for nonfixed and fixed boundary targets in the DOTA dataset, sorted by value.

These consistent gains across both DIOR and DOTA datasets confirm the effectiveness and generalization capability of our framework. The detailed comparison results are presented in Table IV, while Fig. 9 provides visual analysis of performance across different boundary types. The success of DOTA, which features different target distributions and imaging characteristics, further validates that KGCS's structured knowledge injection approach effectively bridges the domain gap in various aerial detection scenarios.

3) Computational Cost and Deployment Efficiency Analysis: The KGCS system forms a structured description dictionary assisted by expert knowledge. Since the semantic descriptions are optimized with expert knowledge, the accuracy differences among models are minimal. Therefore, this section focuses on analyzing the computational costs and deployment efficiency of the description dictionary formation and inference process from three perspectives: token consumption, inference time, and number of inference calls.

The key modules involved in forming the description dictionary are as follows.

1) *One-Time Semantic Dictionary Construction (SDM Module):* For each category, only *3–5 representative images* are selected. Structured semantic descriptions are

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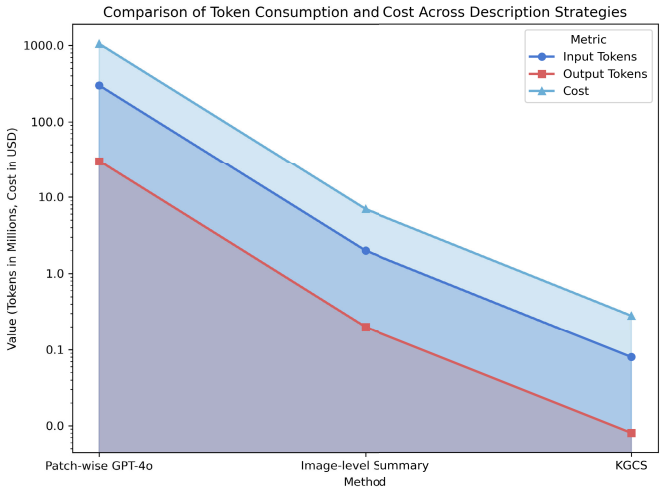


Fig. 10. Comparison of token consumption across description strategies (tokens in millions, cost in USD).

generated by combining GPT-4o with expert knowledge. This dictionary can be reused across datasets, avoiding repetitive textual generation for every image and greatly reducing token consumption. Token cost occurs only once during the construction phase, with no additional cost in later stages.

2) *Structural Proposals and Candidate Compression (OPM Module):* Compared to traditional dense sliding-window methods (processing over a hundred candidate regions per image), the OPM module integrates structural clustering and contour filtering, retaining only *20–50 high-quality candidate regions*. These candidates are designed to reduce the number of matches between description dictionary entries and image regions in the subsequent ISM module, thereby significantly lowering the number of large-model inference calls and reducing inference costs. Additionally, OPM employs *CLIP* for lightweight image–text similarity filtering to quickly discard low-relevance candidates, preserving only structurally clear and semantically relevant regions for downstream large-model inference, further reducing the inference burden.

Taking the DIOR dataset as an example (20 classes, 100 images per class, totaling 2000 images), we compare three semantic description strategies in terms of token-level computational cost under the GPT-4o pricing scheme. GPT-4o charges \$2.5 per million input tokens and \$10 per million output tokens. For a typical 800×800 image, the prompt (input) length is around 1000 tokens, and the generated description (output) is approximately 100 tokens.

1) *Patch-Based GPT-4o Calls:* Each image is processed with approximately 150 region-level prompts (median of 100–200 segments). For 2000 images, this results in around 3 00 000 individual prompt-response pairs, consuming over 300 million tokens in total. The estimated cost exceeds \$1050, with significant latency due to high inference frequency.
2) *Image-Level Summary Generation:* Each image is described via a single prompt. Across 2000 images, the total input and output tokens reach approximately two million and 2 00 000, respectively. The total cost is around \$7.00. While this strategy reduces the number of inference calls, it is often constrained by prompt length, limiting contextual completeness and generalization.
3) *KGCS Approach:* A one-time semantic dictionary is constructed using 3–5 representative images per category (20 categories in total), resulting in approximately 80 000 input and 8000 output tokens. The total token cost is only \$0.28. Once generated, the dictionary is reused across the entire dataset without per-image inference. Token consumption remains fixed regardless of dataset size, and the average per-image latency is reduced to 8.0 s through lightweight region proposals and early filtering.

TABLE V
DEPLOYMENT COMPARISON OF THE LOCAL MODEL (7B, 32B, AND 72B) AND GPT-4O ACROSS KEY METRICS

Metric	7B	32B	72B	GPT-4o
Latency (s/img)	12.0	624.0	715.0	**8.0**
GPU Memory (GB)	16.8	76.8	172.8	**0.0**
Env. Dependency	High			**Low**
Stability	Unstable			**Stable**
Scalability	Limited			**High**

As visualized in Fig. 10, the KGCS approach demonstrates superior cost-efficiency.

1) *Token Reduction:* 300 × less than patch-based methods (0.3M versus 300M tokens).
2) *Cost Savings:* 99.97% cost reduction compared to patch-based (\$0.28 versus \$1050).
3) *Scalability*: Flat cost curve versus linear growth of alternatives.

4) Model Selection and Deployment Cost Analysis: The adoption of GPT-4o in this work is not primarily motivated by accuracy improvements—expert-guided dictionary construction ensures semantic consistency across models—but by deployment efficiency. As detailed in Table V, large-model latency tests (7B/32B/72B) were conducted on an 8× RTX 4080 SUPER (32GB) platform, while all KGCS experiments ran on a single RTX 4070 Ti (12GB).

This configuration conclusively demonstrates GPT-4o's engineering advantages in operational efficiency, delivering two critical benefits for real-world deployment.

1) *Significant Cost Reduction:* Eliminates the need for expensive multi-GPU setups (e.g., 8 × 4080 SUPER \$12 000), enabling deployment on accessible hardware like the RTX 4070 Ti (\$800)—an 85% cost saving while maintaining full functionality.
2) *Predictable Low Latency:* Achieves consistent subsecond inference times (8 s per image) regardless of scene complexity, unlike local models that exhibit extreme latency variance (12–715 s, up to 60× fluctuation).

F. Parameter Settings and Transferability

KGCS is designed to minimize dataset-specific parameter tuning. From an implementation perspective, its parameters

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can be divided into three categories: backbone configurations, inference-related parameters, and a semantic scene dictionary. Among them, only the semantic dictionary requires manual adjustment when transferring to a new dataset.

1) Backbone Configuration: KGCS employs fixed pre-trained models, including SAM (ViT-H) for segmentation and CLIP ViT-L/14@336px for vision–language matching. All models are used in a frozen manner, without task-specific fine-tuning. Therefore, no training hyperparameters are involved, and the same backbone configuration is directly reused across datasets. Owing to the modular design of KGCS, these backbone models can be directly replaced by newer versions within the same model families (e.g., updated SAM or CLIP variants) without modifying the overall framework or parameter settings, enabling long-term extensibility.

2) Inference Parameters: Parameters related to object scale, spatial structure, and proposal generation are not manually specified. They are automatically determined by the OPM based on the geometric statistics of segmentation masks [(5)–(8)]. As a result, these parameters adapt to different resolutions and object distributions without dataset-specific tuning.

3) Semantic Scene Dictionary: The semantic scene dictionary is the only dataset-dependent component in KGCS and is constructed in a class-specific manner. For each target class, the dictionary contains 3–5 concise textual descriptions provided by users or generated by a large language model. One description corresponds to the target object, while the remaining entries describe the most salient distinguishing characteristics of typical co-occurring objects in the same scene.

All descriptions are directly compared with candidate regions through the vision–language model, and target confidence is determined by relative similarity between the target description and its corresponding distractor descriptions. This design does not expand the category space and does not involve negative supervision.

When applying KGCS to a new dataset, users only need to update the dictionary entries if the scene composition changes. The dictionary structure, backbone models, and inference procedures remain unchanged.

G. Ablation Study

To quantitatively evaluate the contribution of each component in the KGCS framework, we conduct a comprehensive ablation study under the GZSD setting on the DIOR dataset. The analysis is organized at two levels: the *module level* assesses the impact of the core modules (OPM, SDM, and ISM) on the overall pipeline, while the *key component level* examines the necessity and interactions of the scene description dictionary and the two-stage adaptive filtering mechanism—the semantic core of the system.

1) Module-Level Analysis: We decompose the complete framework into three progressive configurations to isolate the contribution of each module. Table VI reports the performance changes as modules are incrementally added.

Introducing the OPM module improves recall from 31.6% to 32.8% (+1.2%), demonstrating its ability to generate additional candidate regions, particularly for boundary-ambiguous targets. However, mAP drops from 4.2% to 3.1% (−1.1%), showing that without semantic refinement, the enlarged candidate set introduces excessive noise, reducing overall precision.

TABLE VI
MODULE-WISE PERFORMANCE IMPACT ON THE DIOR TEST SET (GZSD)

Base	OPM	SDM&ISM	mAP	Recall
✓			4.2	31.6
✓	✓		3.1	32.8
✓	✓	✓	**15.4**	**48.8**

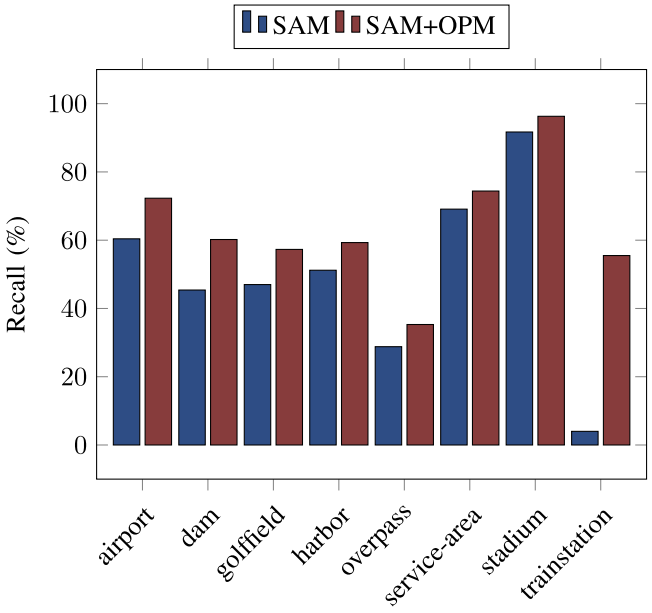


Fig. 11. Recall improvement for boundary-ambiguous categories using SAM and SAM+OPM.

The capability of OPM to enhance recall for boundary-ambiguous categories is further validated in Fig. 11. For structurally complex categories such as *trainstation*, recall increases dramatically from 4.0% to 55.5%. Similar improvements are observed for *dam* and *airport*, confirming that OPM's dual-path strategy effectively expands candidate coverage.

However, due to CLIP's limited discriminative capacity at fine spatial scales, many newly introduced candidate regions cannot be reliably classified, causing a decline in precision. This highlights the necessity of semantic-level refinement—specifically, the SDM and ISM modules—to separate informative regions from distractors.

Integrating SDM and ISM yields substantial gains in both mAP and recall, confirming the effectiveness of the semantic refinement framework. Its improvements stem from two closely coordinated components.

1) *Discriminative Scene Description Dictionary:* Combining domain knowledge with LLM-generated priors, it constructs a semantic repository containing distinctive target attributes and common distractors. This enhances

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TABLE VII
ABLATION ON EXPERT-KNOWLEDGE GRAPH AND TWO-STAGE ADAPTIVE FILTERING

Configuration	mAP (%)	Recall@100 (%)
Base model + vanilla CLIP	4.2	31.6
+ Expert-Knowledge Graph (EKG)	9.6	53.5
+ Two-Stage Adaptive Filtering	1.6	50.6
EKG + Two-Stage Adaptive Filtering	**15.4**	**48.8**

CLIP's fine-grained alignment ability and improves interclass discrimination.

2) *Two-Stage Adaptive Filtering:* Guided by structural priors, the first stage dynamically calibrates thresholds based on confidence statistics for initial screening, while the second stage incorporates boundary-aware retention ratios as a robust safeguard. Together, they suppress low-quality proposals while preserving recall, optimizing the precision–recall balance.

This demonstrates that, under strict zero-annotation constraints, the combination of *high-quality semantic priors* and *data-driven adaptive filtering* forms an effective pathway for improving the domain adaptability of foundation models.

2) Key Component-Level Analysis: Building on the validated module-level design, we further analyze the two most critical semantic components: the scene description dictionary and the two-stage adaptive filtering mechanism. Table VII presents an ablation study illustrating their respective contributions and synergistic effects.

The baseline with a simple template dictionary performs poorly. Incorporating the expert-enhanced scene description dictionary significantly boosts both mAP and recall to 9.6% and 53.5%, respectively, demonstrating that high-quality semantic priors are fundamental to improving zero-shot performance. In contrast, applying only the second-stage filtering mechanism to a weak baseline yields high recall (50.6%) but causes mAP to collapse to 1.6%, confirming the *quality before quantity* principle: without reliable semantic alignment, filtering alone cannot recover precision.

Sequentially integrating both components—first obtaining discriminative matching scores via the dictionary, then applying the two-stage adaptive filtering—raises mAP to 15.4% while maintaining strong recall (48.8%). This shows that the filtering mechanism effectively leverages high-quality confidence distributions to balance precision and recall. Hence, the scene description dictionary serves as the primary performance driver, while the filtering mechanism functions as a critical optimizer; both are complementary and must be applied in order, forming the foundation of KGCS for zero-annotation detection.

Finally, to illustrate our evaluation protocol, Fig. 12 provides a visual comparison of detection results, where green denotes the proposed model, red the baseline, and blue the ground truth.

H. Strategic Interpretation of the Recall–mAP Tradeoff

The 22.7% improvement in Recall@100 on the DOTA dataset, alongside a modest 0.6%–0.7% increase in mAP, should not be interpreted as a limitation of KGCS. Instead, it reflects a deliberate operating strategy formulated under stringent resource constraints. This design principle is guided by three key considerations:

1) *Resolution Limitations of CLIP:* Zero-shot CLIP exhibits constrained interclass discrimination in aerial imagery. Adjusting the similarity threshold introduces an inherent tradeoff: increasing the threshold suppresses false positives and raises mAP at the cost of recall, whereas lowering it boosts recall while reducing precision. Given this bounded discriminative capacity, we intentionally bias the operating point toward higher recall, thereby preserving a broader set of candidate regions for lightweight downstream refinement (e.g., a single global re-ranking step using GPT-4o).
2) *Resource-Aware Design Objectives:* Our study targets a deployment context characterized by zero manual annotation, single-GPU execution, and an inference latency of no more than 8 s per image. Under these constraints, prioritizing recall while maintaining reasonable mAP offers several advantages: i) complete removal of labeling overhead; ii) reduced memory and computational cost suitable for edge devices; and iii) full reproducibility on commodity hardware.
3) *Two-Stage Recall–Then–Filter Paradigm:* KGCS adopts a "recall-first, filter-later" strategy: Stage 1 (OPM + SDM) aims to maximize candidate coverage, whereas Stage 2 (the two-stage adaptive filtering in ISM) conducts confidence-based and structure-aware refinement. As demonstrated in Table VI, once candidate scores are recalibrated via the scene description dictionary, the proposed adaptive filtering mechanism increases precision without significantly compromising recall. Consequently, the observed DOTA performance profile results from intentional system-level decisions rather than algorithmic limitations.

In summary, the recall–mAP characteristics of KGCS arise from a conscious tradeoff tailored to strict resource conditions. This positions KGCS as a practical, annotation-free alternative to resource-intensive multimodal alignment methods that rely on large labeled datasets, extensive computational budgets, or prolonged fine-tuning.

I. Limitations Analysis

KGCS is designed as a zero-annotation framework for semantic region classification in aerial imagery, eliminating the need for large-scale annotations or fine-tuning. This paradigm is especially suitable for scenarios with limited training data or constrained computational resources, but it also introduces inherent tradeoffs that shape the system's capabilities.

1) CLIP Constraints and Expert Knowledge Compensation: Although CLIP provides strong open-vocabulary vision–language alignment, its coarse-grained discriminative capacity limits its ability to differentiate visually subtle or structurally similar categories (e.g., train stations versus residential blocks and fighter jets versus transport aircraft) when used without task-specific fine-tuning. This constraint is not a

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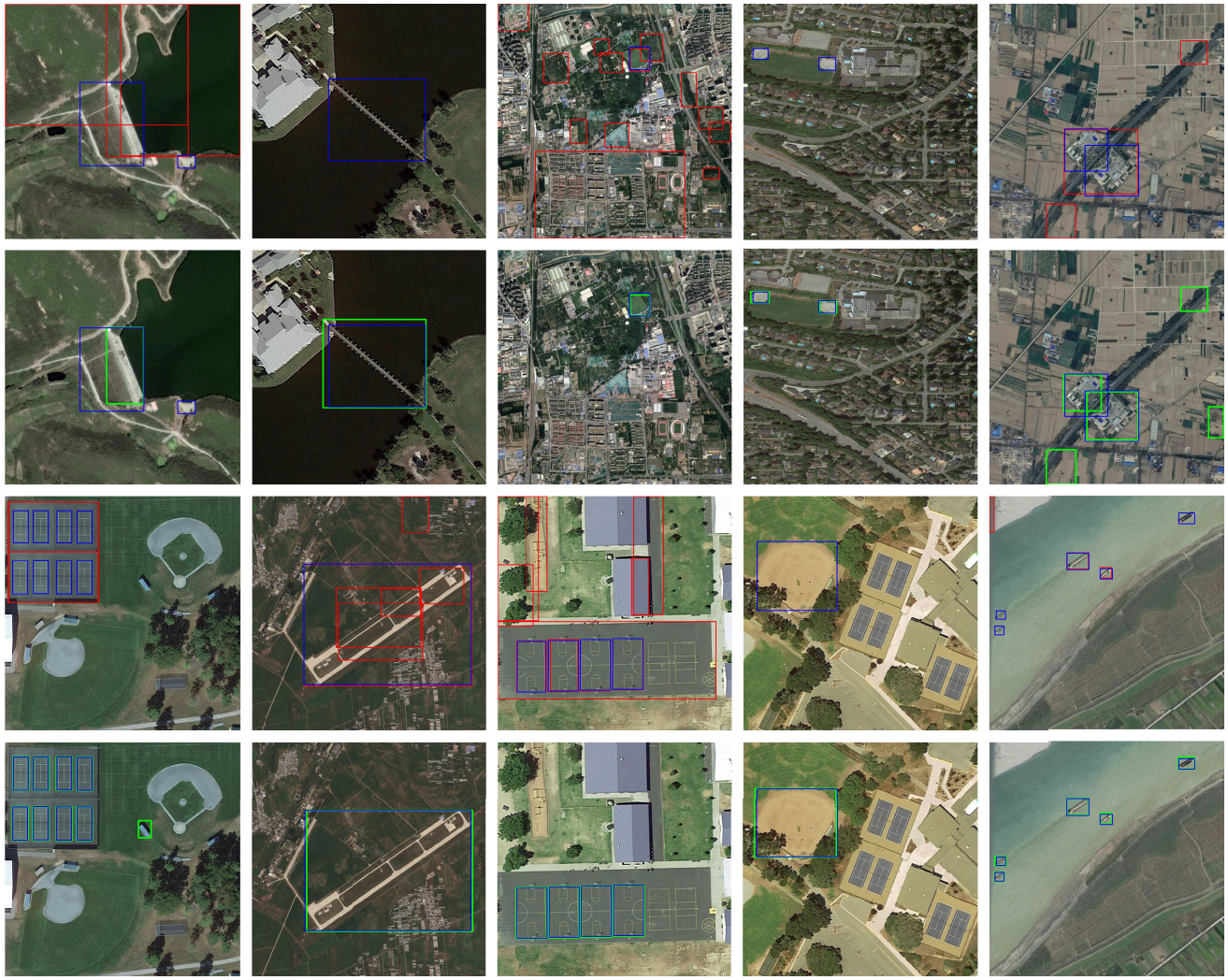


Fig. 12. Visual comparison of detection outcomes. Green: proposed model; red: baseline; blue: ground truth.

limitation of KGCS itself, but an inherent characteristic shared by all zero-annotation systems relying on frozen VL models.

To address this, KGCS integrates an expert-driven semantic description dictionary that encodes structural and contextual priors to compensate for CLIP's perceptual gaps. Although dictionary construction introduces a one-time human effort, it is significantly lighter than dataset annotation and dramatically improves discriminative performance in data-sparse remote sensing environments.

2) Efficiency and Resource Considerations: Compared with region-wise classification pipelines requiring multiple LLM queries (e.g., GPT-4o), KGCS greatly reduces inference cost by performing only a single global reasoning step per image. Nevertheless, two deployment challenges remain.

1) *Inference Latency:* The current end-to-end inference time is approximately 8 seconds per 2048 $\times$ 2048 image, which is suitable for batch processing but still suboptimal for real-time applications.
2) *Memory Usage:* Processing high-resolution aerial imagery typically requires 8—9 GB of GPU memory, which may limit deployment on extremely resource-constrained devices.

Future improvements may explore lightweight ViT-based LLMs, prompt compression, and on-device distillation to further improve throughput and make KGCS more compatible with lower-end edge platforms.

IV. Conclusion

This article presents KGCS, a novel zero-annotation framework for aerial remote sensing detection that addresses key challenges—including structural ambiguity, semantic discrimination difficulty, and annotation scarcity—by systematically integrating domain expert knowledge with foundation vision–language models. To this end, KGCS decomposes the detection pipeline into three cooperative modules: the OPM, SDM, and ISM, which jointly enable complementary enhancements at the structural, semantic, and alignment levels.

Specifically, OPM improves the completeness and accuracy of candidate regions for structurally complex or boundary-ambiguous targets using a structure-aware mask clustering and adaptive sliding window strategy. SDM introduces an expert-guided hierarchical description dictionary that compensates for CLIP's coarse-grained limitations and enhances fine-grained semantic discrimination. ISM further refines the recognition results through confidence calibration and two-stage adaptive

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filtering, effectively suppressing low-quality predictions and reducing false positives.

Extensive experiments on multiple public aerial remote sensing datasets demonstrate that KGCS yields substantial improvements in recall for ambiguous targets while reducing false detections. Moreover, KGCS requires no manual annotation, incurs low inference cost, and achieves inference speeds suitable for deployment on *lightweight GPU-based edge nodes*.

Future work will focus on enhancing mask quality in complex scenes and improving multimodal cross-domain generalization, aiming to further advance the performance and applicability of KGCS.

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